

Module 16: Capstone Systems Synthesis - Practice Quiz

Use this low-stakes quiz after reviewing the module keys and learning questions.

1. A strong biological explanation that spans scales does what?

- A. stays at a single scale only
- B. avoids using any evidence
- C. connects mechanisms from molecules up to ecosystems into one account
- D. relies on vocabulary alone
- Answer: C
- Why: Synthesis links causes across scales, from molecular to ecological, into one coherent, evidence-backed explanation.

2. When rising body temperature makes you sweat and cool back down, that is an example of:

- A. a negative feedback loop that reduces change
- B. a random mutation
- C. a tradeoff with no feedback
- D. genetic drift
- Answer: A
- Why: Negative feedback counteracts a change to keep conditions stable, which is a core systems idea behind homeostasis.

3. "A larger body helps an animal retain heat but also requires more food." This statement describes a:

- A. feedback loop
- B. mutation
- C. species concept
- D. tradeoff
- Answer: D

- Why: A tradeoff pairs a benefit in one function with a cost in another, a common pattern in biological systems.

4. To support a claim that spans several biological scales, the strongest evidence is:

- A. a single memorable anecdote
- B. an evidence chain in which each step's evidence matches its scale
- C. a list of vocabulary words
- D. personal opinion alone
- Answer: B
- Why: An evidence chain links the claim to appropriate data at each scale, which makes a cross-scale explanation testable and convincing.